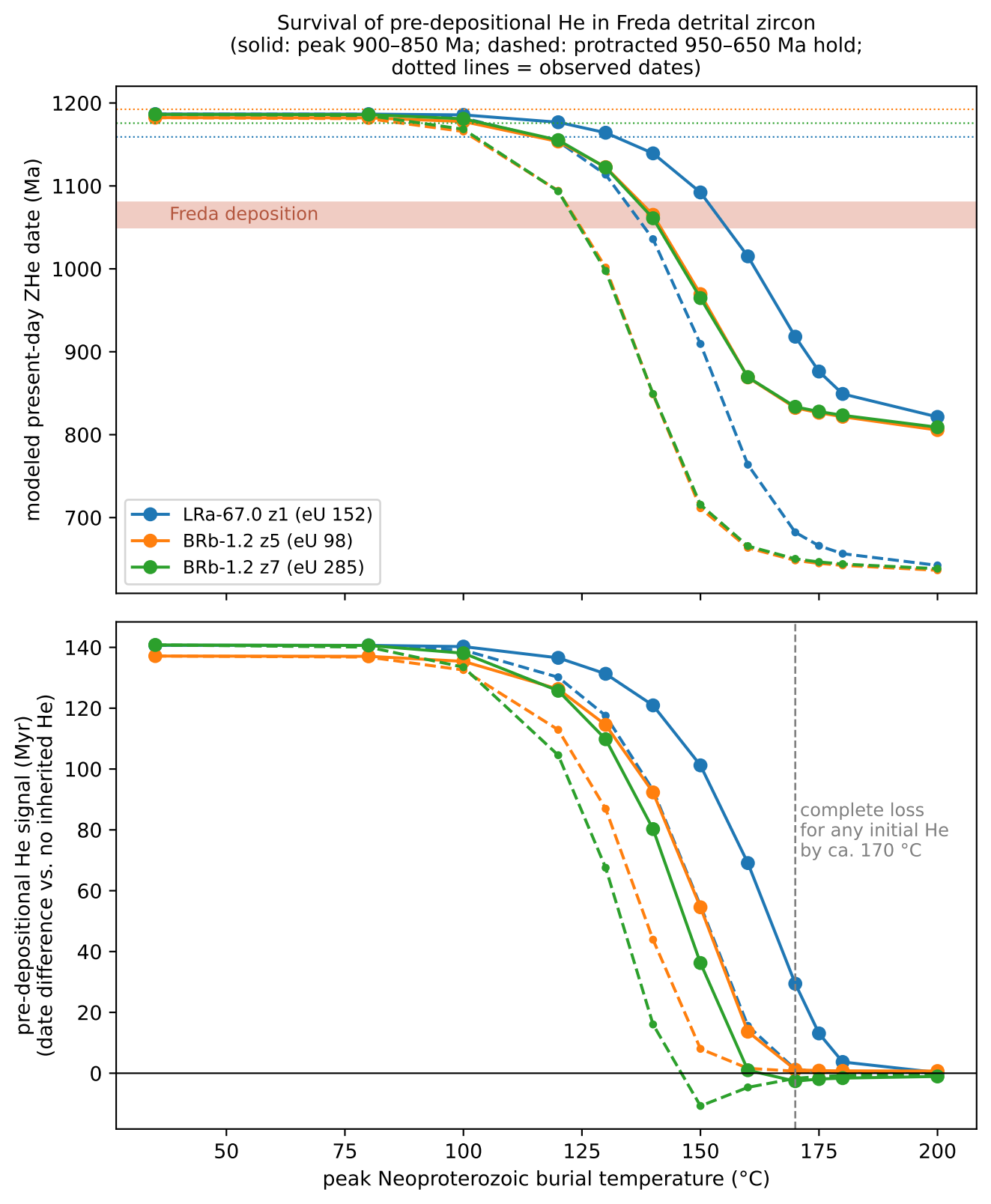
Survival of pre-depositional He in Freda detrital zircon
(solid: peak 900–850 Ma; dashed: protracted 950–650 Ma hold;
dotted lines = observed dates)
modeled present-day ZHe date (Ma)
1200
1100
1000
900
800
700
Freda deposition
LRa-67.0 z1 (eU 152)
BRb-1.2 z5 (eU 98)
BRb-1.2 z7 (eU 285)
pre-depositional He signal (Myr)
(date difference vs. no inherited He)
140
120
100
80
60
40
20
0
complete loss
for any initial He
by ca. 170 °C
50
75
100
125
150
175
200
peak Neoproterozoic burial temperature (°C)